

Raafi Riyaz

MSc AI | [Stanford University Online ML Certified](#)

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EXPERIENCE

RL Environment Engineer (contract)

April 2026 – Present

Tensium

London, UK

- Created long-horizon reinforcement-learning environments in Harbour format with 70–150 causal steps for frontier labs, maintaining a **25% pass@8 performance gap** between weaker and frontier models to ensure meaningful task discrimination.
- Used the **HUD platform** to design, debug, and analyze RL environments, engineering tasks resistant to oracle leakage, reward hacking, public-data overfitting, and shortcut-based solutions while targeting challenging long-horizon reasoning and coding behavior.
- Built an automated **PR-triggered long-horizon RL QA pipeline** with GitHub Actions and an internal harness, running isolated environment/grader checks and surfacing automated pass/block verdicts to replace manual soundness review, reviewed and successfully delivered 4 tasks in a 20 hour sprint.

Machine Learning Engineer (contract)

Oct. 2025 – Present

Mercor, USA

London, UK (Remote)

- **Generalist AI Expert** evaluated **frontier AI models** to identify performance gaps and contribute to capability improvements.
- Generated human baselines for **SWE and Terminal Efficiency benchmarks** using only **official Python documentation**, replicating pre-GPT environments to produce high-quality evaluation and training data.
- **Rubrics Academy Fellow** tackled complex **prompt reasoning and technical problems** through first-principles architectural design under strict constraints on AI tool usage.
- **Project Launchpad** contributed to **Frontier AI Research's** expansion of **OpenAI's MLE-bench** by fixing LLM-generated ML code, creating plan-code post-training data, and merging **30+ pull requests** across task conversions, datasets, and evaluation improvements using **AIRA-dojo** on **10 AWS H100 clusters**.
- **Project Vulcan** collaborated with **frontier lab** to extend **OpenAI's MLE-bench** into a comprehensive benchmark by transforming **Kaggle competitions** into reproducible, **Docker-based evaluation tasks** and adding recent **NeurIPS, ICML, and ICLR** datasets across **computer vision, NLP, time-series, and tabular domains**.

Founder and AI Engineer

Sep. 2025 – Present

VideoVoice & FeedHire

London, UK (Remote)

VideoVoice - AI Video Translation & Dubbing

- **Research-engineering the limits of fully open-source video/voice pipelines** - zero closed-source APIs, with all models self-hosted on a total of just **5 GPUs**.
- Built an AI video translation system with **zero-shot voice cloning** via a **6-step pipeline** (**FFmpeg** → **OpenAI Whisper** → **open LLMs** → **Chatterbox/OmniVoice/Qwen TTS** → **time-sync** → **merge**) supporting **600+ languages**, with dubbing quality surpassing YouTube/Instagram auto-dubs (live demos on site).
- Built **DramaBox**, a text-to-reel generator turning any script into a captioned vertical video via a **5-stage pipeline** (**text** → **AI scene direction** → **emotionally-expressive TTS** → **scene-context image generation** → **rendered reel**) with **word-by-word synced captions**, exportable for **TikTok/Reels/Shorts** and fully re-runnable per layer (audio, prompts, images).
- Architected scalable **FastAPI** infrastructure on **AWS** and **Hugging Face Spaces GPUs**, exposing **REST + UI endpoints** with **CI/CD** via **GitHub Actions**.

FeedHire - AI Job Data Extraction

- Built a **Dockerized FastAPI microservices** backend with **delta-detection scrapers** for **Telegram, Reddit, and GitHub**, enabling high-frequency, independently scalable AI data extraction; grew to **2000+ global traffic** and **200+ registered users**.

- Engineered a multi-stage extraction pipeline with a **fallback chain (Groq, AWS Bedrock, Ollama)** for high-accuracy job-property extraction resilient to provider failures and rate limits.
- Automated a “**Gold Standard**” **dataset generator** (raw/structured pairs) for model distillation, and integrated **LinkedIn/Discord APIs** for AI-generated promotional posts.

Founding ML Research Engineer

Mar. 2025 – Nov. 2025

Curify-Ai, China

London, UK (Remote)

- Implemented end-to-end **video translation pipelines** spanning WhisperX transcription, translation, multilingual voice cloning, and lip-sync, optimizing multi-stage workflows to reduce processing time
- Developed **temporal audio-visual alignment** using GPT-based post-editing to resolve audio overlap and improve synchronization; deployed ChatterBox TTS, WhisperX, and PaddleOCR as **Dockerized FastAPI microservices**
- Built scalable backend infrastructure with **PostgreSQL, queue-based job orchestration, credit/subscription systems, and REST APIs** for production ML workloads

AI/ML Research Intern

Jul. 2024 – Oct. 2024

City, University of London, UK

London, UK

- Integrated clinical and phylogenetic data to enhance ML models for predicting lung cancer patient survival using a novel dataset with no prior research
- Experimented with survival model techniques and feature engineering to improve patient survival time predictions

AI Engineer

Sep. 2022 – Sep. 2023

Webomates, USA

Mumbai, India (remote)

- Deployed and monitored machine learning models on **AWS EC2**, leveraging advanced **Linux** system capabilities
- Designed and implemented a novel approach using **AWS SQS** to replace Flask-based request handling, leading to team-wide adoption and **30-50% increase in efficiency**
- Developed a multi-modal ML pipeline to detect feature changes in web pages using HTML, user console logs, and visual data.
- Implemented a hybrid model combining XGBoost for HTML data, CNN for image data, and Random Forest for final feature detection, **improving accuracy by 25%**
- Designed Flask application leveraging OpenAI API to generate and enhance test cases for TestOps

Machine Learning Engineer Intern

Oct. 2021 – Jan. 2022

ResoluteAI, India

Mumbai, India (remote)

- Worked on a U-NET Neural Network model architecture to detect defects in fabric videos, handling video-to-frame conversion, image augmentation, and model training/testing
- Led a team of 4 interns on image annotation tasks with OpenCV
- Extracted regions of interest (ROI) and labeled objects using the Canny edge detection algorithm

EDUCATION

MSc Artificial Intelligence (Grade: Merit)

Sep. 2023 – Oct. 2024

City, University of London

London, UK

BSc Information Technology (Grade: Distinction)

Jul. 2019 – Mar. 2021

University of Mumbai

Mumbai, India

PROJECTS

- [Fine-tuned Meta-Llama-3-8B](#) using LoRA on custom dataset, released on Hugging Face with **200+ downloads**.
- Implemented [Super Resolution Residual Network \(SRResNet\)](#) and [Super-Resolution Generative Adversarial Network \(SRGAN\)](#) to enhance image resolution, proposing new improvements to the models, achieving better performance on benchmarks. Also presented first baselines for the Fréchet Inception Distance (FID) following their work.
- Thesis-based [internship research project](#) on predicting cancer patient survival times using clinical and genetic data, applying linear and non-linear regression methods, with a publication in progress.

OPEN-SOURCE CONTRIBUTION

- **Contributor, Exo (40k+ stars):** Architected multimodal support by integrating **Vision-Language Models**.
- **Contributor, MuseTalk (5k+ stars):** Added **Docker** deployment support and **FastAPI** serving.
- **Technical Author, Medium:** Articles revolving around AI/ML space and JEPA world model.
- **Guest Speaker:** Introduction to RL session for final-year BTech students at JMI University with Bytes Classes.
- **Technical Author:** Authored several articles for AI in software testing domain for Webomates, USA.

AWARDS & ACHIEVEMENTS

- **Technical AI Safety, BlueDot:** Completed application-based AI safety course (alignment, control). **(May 2026)**
- **RL fellowship:** Offered by General Reasoning; led by Llama lead and, Papers with Code founder **(Mar 2026)**
- **AIUK 2025 Specialist:** Invited to present at UK national event for mental health AI contributions. **(Mar 2025)**
- **Winner of Harmony AI Challenge:** Winner in collaboration with UCL and 3 other universities. **(Jan 2025)**
- **Co-Founder, Feedhire:** Scaled the platform to **1500+ global users** traffic in 6 months. **(Dec 2025)**
- **ML Researcher:** Selected for an internship on applying machine learning to lung cancer research. **(Oct 2024)**
- **Early Career Offer:** Promoted within 3 months of a 6-month internship at Webomates, USA. **(Jan 2023)**